

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND ENGINEERING (IOT)

VII SEMESTER

R22

VII SEMESTER		R22				
Course Code	Title of the Course	L	T	P/D	CH	C
22PC1IN401	Embedded GPU	3	0	0	3	3
22PC1AM402	DevOps	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CS402	Deep Learning	3	0	0	3	3
22PE1DS402	Distributed Computing					
22PE1IN401	Communication Technologies for IoT					
22PE1CS304	Cyber Security					
22PE1IN402	IoT for Smart Cities					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Machine Learning Stream	3	0	0	3	3
	Computer Science Stream					
	Networks Stream					
	Software Engineering Stream					
	Cyber Security Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2IN401	Embedded GPU Laboratory	0	0	2	2	1
22PC2AM402	DevOps Laboratory	0	0	2	2	1
22PW4IN401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
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VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PC1AM401	Image Processing and Computer Vision	3	0	0	3	3
22PE1IN403	Wearable Computing					
22PE1CS404	Augmented Reality and Virtual Reality					
22PC1CY401	Block chain Technology					
22PE1IN404	IoT System Design					
	PROFESSIONAL ELECTIVE – VI					
22PE1IN405	5G and IoT Technologies	3	0	0	3	3
22PE1IN406	Industrial IoT 4.0					
22PE1DS406	Quantum Computing					
22PE1IN407	Edge and Fog Computing					
22PE1IN408	Privacy and Security in IoT					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4IN402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

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VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1IN401) EMBEDDED GPU

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of GPU architectures and optimization for embedded GPUs
- To gain knowledge on memory handling techniques with CUDA
- To write programs for massively parallel processors
- To understand the issues in mapping algorithms for GPUs
- To introduce different GPU programming models

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the GPU computing architecture

CO-2: Interpret code with GPU programming environments

CO-3: Design and develop programs using GPU processing power

CO-4: Evaluate GPU algorithm application using NVIDIA Nano

CO-5: Develop solutions to solve computationally intensive problems in various fields

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	-	-	-	2	2	2	2	2	2	2
CO-2	3	2	2	2	3	-	-	-	2	2	2	2	2	2	2
CO-3	3	3	3	2	3	-	2	2	3	3	3	3	2	2	2
CO-4	3	3	3	3	3	-	2	2	3	3	3	3	2	2	2
CO-5	3	3	3	3	3	1	2	2	3	3	3	3	2	2	2

UNIT-I:

Introduction to Embedded GPU: Review of Traditional Computer Architecture – Basic five stage RISC Pipeline, Cache Memory, Register File, SIMD instructions, Evolution, GPU Computing. Embedded v/s discrete GPUs, Performance v/s Power tradeoff in embedded GPUs; Understanding Parallelism with GPU –Typical GPU Architecture - CUDA Hardware Overview - Threads, Blocks, Grids, Warps, Scheduling.

UNIT-II:

Memory Handling with CUDA: Shared Memory, Global Memory, Constant Memory and Texture Memory, Unified memory in embedded GPUs. Optimizations in embedded GPUs: Unified memory, reduced peripherals, cost reductions, performance, higher operating temperature range, longer support timelines.

UNIT-III:

GPU Programming & Execution Models: Execution model of GPU- memory allocation and data transfer to DRAM by CPU, kernel launch, execution by threads. Using CUDA - Multi GPU - Multi GPU Solutions - Optimizing CUDA Applications: Problem Decomposition, Memory Considerations, Transfers, Thread Usage, Resource Contentions, Self-tuning Applications.

UNIT-IV:

Programming Issues

Common Problems: CUDA Error Handling, Parallel Programming Issues, Synchronization, Algorithmic Issues, Finding and Avoiding Errors. Profiling and optimizing

UNIT-V:

Algorithms on GPU: GPU DEVICES AND APPLICATIONS, Parallel Patterns: Convolution, Prefix Sum, Sparse Matrix - Matrix Multiplication - Programming Heterogeneous Cluster - CUDA Dynamic Parallelism. Example Deep Neural Network which uses above algorithms, Introduction to OpenCL – OpenCL Device Architectures – Basic OpenCL – examples – Understanding OpenCL
Introduction to Jetson Nano by NVIDIA, Application design on Jetson Nano

TEXT BOOKS:

1. The World of Scripting Languages, David Barren, Wiley
2. Learning PHP, MySQL, JavaScript, CSS&HTML5: A Step-by Step Guide to Creating Dynamic Websites, 3rd Edition, O'Reilly
3. Professional PHP Programming, Jesus M. Castagnetto, Harish Rawat, Deepak T. Veliath, Wrox Publication

REFERENCES:

1. Perl by Example, E. Quigley, Pearson Education
2. Programming Perl, Larry Wall, T. Christiansen and J. Orwant, O'Reilly
3. Open-source Web Development with LAMP using Linux Apache, MySQL, Perl and PHP, J. Lee and B. Ware, Addison Wesley, Pearson Education

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1AM402) DEVOPS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce Agile methodologies and DevOps principles, highlighting their role in improving software development efficiency and addressing bottlenecks
- To explore modern software architectures, emphasizing the transition from monolithic to microservices and serverless models while integrating DevOps practices
- To develop understanding in version control and containerization, utilizing Git, GitHub, Docker, and Kubernetes for efficient project collaboration and deployment
- To familiarize with CI/CD and automation skills, covering Jenkins, CircleCI, GitHub Actions, infrastructure as code (Terraform), testing and monitoring tools
- To introduce DevSecOps and security best practices, focusing on secure coding, secret management, container security, and automated security testing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse and apply DevOps principles to enhance business agility, and address software development bottlenecks

CO-2: Design and manage software architectures, including monolithic, microservices and serverless using DevOps practices

CO-3: Implement version control and collaboration using Git and GitHub, and manage containerized applications using Docker and Kubernetes

CO-4: Build and manage CI/CD pipelines using Jenkins, CircleCI, and Github Actions for automated deployments and testing

CO-5: Understand IaC and the need to integrate security in DevOps with DevSecOps and GitOps to ensure secure and efficient software delivery

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	2	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	2	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

UNIT-I:

Introduction to DevOps & Business Agility: Introduction to DevOps (What is DevOps, benefits and goals of DevOps, DevOps lifecycle and phases), DevOps and Business Agility (What is business agility, DevOps's role in enhancing agility, collaboration

between dev, QA and ops), Bottlenecks (Identifying bottlenecks in software delivery, common bottlenecks, how DevOps addresses bottlenecks).

UNIT-II:

Software Architectures & the Need for DevOps: Introduction to Agile (Agile vs Waterfall model, Agile methodologies – Scrum, Kanban, Continuous Integration and Continuous Development – CI/CD overview, how Agile and DevOps work together), Software Architectures (Importance of separation of concerns, different types of architectures and examples, Monolithic vs Microservices vs Serverless, benefits and challenges of each)

UNIT-III:

Project Management with Version Control, Containerization and Kubernetes: Version Control Systems (Introduction to VCS, Types of VCS, Git vs other VCS tools, Important Git commands, GitHub for collaboration – branching and merging, pull requests, code reviews), Containerization with Docker (What are containers, Docker vs Virtual Machines, building and running containers, port mapping, container networking overview, DockerHub introduction), Kubernetes (What is Kubernetes, core concepts- pods, nodes, deployments, services, Configmaps and secrets, Kubernetes networking overview, scaling applications (horizontal, vertical), persistent storage, deploying applications- kubectl, config files)

UNIT-IV:

CI/CD, Testing and Monitoring: Importance of CI/CD, Jenkins overview (pipelines, build dependencies, plugins, triggers), CI/CD with CircleCI, Using Github Actions for CI/CD, Testing and Automation (importance of automated testing, types of testing – unit, integration, end-to-end, Test Driven Development (TDD), API testing using Postman, JUnits/PyTests), Monitoring and Logging (Importance of logging and monitoring in DevOps, monitoring tools – Prometheus, Grafana, logging tools – Splunk, New Relic, implementing alerts and logging analysis)

UNIT-V:

Infrastructure as Code(IaC), GitOps and DevSecOps Overview: What is IaC and Terraform, writing Terraform configurations, Terraform State Management and Kubernetes infrastructure, GitOps and ArgoCD (brief introduction to GitOps, ArgoCD setup and declarative deployments), DevSecOps – Security in DevOps (Shift-left security & container image scanning, secret management – vault, AWS/Azure secret manager, Automating security in CI/CD)

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, 2018
2. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint, Wiley

REFERENCE:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CS402) DEEP LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To gain a grasp of the fundamental concepts of deep learning and regularization in the context of deep learning
- To examine approaches for optimization and creating high-performing deep learning networks while taking into account hyper parameters
- To gain understanding of recurrent and recursive networks
- To develop effective solutions for real-world problems utilizing Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Acquire a comprehensive understanding of the fundamental principles underlying Deep Learning

CO-2: Comprehend and implement regularization techniques for neural networks in deep learning

CO-3: Apply optimization techniques to train deep models effectively

CO-4: Acquire a foundational understanding of Convolutional Neural Networks (CNNs), including their basic principles and architecture

CO-5: Investigate various architectures of Recurrent Neural Networks (RNNs) to explore their diverse capabilities and applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	1	1	2	1	1	-	3	2
CO-2	3	3	1	2	1	2	1	1	1	1	1	1	-	3	2
CO-3	3	3	1	2	1	2	1	1	1	1	1	1	-	3	2
CO-4	3	3	1	2	1	2	1	1	1	1	1	1	-	3	2
CO-5	3	1	2	1	2	1	1	1	1	1	1	1	-	2	2

UNIT-I:

Introduction to Deep Learning: Biological neuron, Importance and Applications of Deep Learning. Issues of Deep Learning, Supervised Learning Networks, Perceptron Networks.

Deep Feedforward Networks: Historical Trends in Deep Learning Deep Feed forward Networks: Example: XOR, Gradient-based learning, Hidden units.

UNIT-II:

Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization under constrained problems, Dataset Augmentation, Noise Robustness, Semi-Supervised Learning, Multi-Task Learning, Early

Stopping, Parameter Tying and Sharing, Sparse Representations, Bagging and Ensemble Methods, Dropout.

UNIT-III:

Optimization for Training Deep Models: How learning differs from Pure Optimization, Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies,

UNIT-IV:

Convolutional Networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function.

UNIT-V:

Sequence Modelling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks-Teacher forcing and networks with output recurrence, Computing the gradient in a recurrent neural network, recurrent networks as directed graphical models, LSTM, Deep Recurrent Networks, Recursive Neural Networks.

TEXT BOOKS:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press, 2016
2. Neural Networks and Deep Learning, Charu C. Aggarwal, Springer, 2018

REFERENCES:

1. Deep Learning Methods and Applications, Foundations and Trends® in Signal Processing, Li Deng and Dong Yu, Volume 7, Issues 3-4
2. Deep Learning Made Easy with R - A Gentle Introduction for Data Science, Dr. N. D. Lewis, Create Space Independent Publishing Platform, 2016
3. Manning Early Access Program Deep Learning with R Version 1, François Chollet, J. J. Allaire, MEAP Edition, Manning Publications, 2017

ONLINE RESOURCES:

1. <https://www.coursera.org/learn/neural-networks-deep-learning>

B.Tech. VII Semester

(22PE1DS402) DISTRIBUTED COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce concepts related to distributed computing systems
- To focus on performance and flexibility issues related to systems design decisions
- To expose students to current literature in distributed systems
- To prepare students for a real-world distributed application design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand models of distributed computing

CO-2: Analyze algorithms for coordination, communication and synchronization in distributed systems

CO-3: Analyze distributed shared memory models

CO-4: Design and implement distributed file systems

CO-5: Design distributed algorithms for handling deadlocks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	-	-	-	-	-	-	-	1	1	-	-
CO-2	2	2	1	1	1	-	-	-	-	-	-	-	-	1	-
CO-3	3	3	2	2	2	-	-	-	-	-	-	-	1	-	-
CO-4	1	1	3	3	2	-	-	-	-	-	1	1	-	-	-
CO-5	-	-	1	1	2	-	-	-	-	-	-	2	-	-	-

UNIT-I:

Distributed Computing Introduction: Definition, Relation to parallel systems, synchronous vs asynchronous execution, design issues and challenges .A Model of Distributed Computations: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Logical Time: Logical clocks, scalar time, vector time, Matrix time, virtual time, Physical clock synchronization – NTP.

UNIT-II:

Global State and Snapshot Recording Algorithms: System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

Message Ordering and Group Communication: Message ordering paradigms, Group communication, Causal order (CO), Total order, Multicast, Propagation trees for

multicast, Application-level multicast algorithms, Fault-tolerant group communication, Multicast algorithms at the network layer.

UNIT-III:

Termination Detection: System model of a distributed computation, Termination detection using distributed snapshots, weight throwing and spanning-tree-based algorithms, Message-optimal termination detection, and Termination detection in a general distributed computing model, Termination detection in the atomic computation model, Termination detection in a faulty distributed system.

Distributed Mutual Exclusion Algorithms: Lamport's algorithm, Ricart–Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal–El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki–Kasami's broadcast algorithm, Raymond's tree-based algorithm.

UNIT-IV:

Deadlock Detection in Distributed Systems: System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy–Misra–Haas algorithm for the AND model, Chandy–Misra–Haas algorithm for the OR model, Kshemkalyani–Singhal algorithm for the P-out-of-Q model .

Distributed Shared Memory: Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, Register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects

UNIT-V:

Check Pointing and Rollback Recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated check pointing algorithm, Juang–Venkatesan algorithm for asynchronous check pointing and recovery, Manivannan–Singhal quasi-synchronous check pointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol.

TEXT BOOKS:

1. Distributed Computing, Ajay D. Kshemakalyani, Mukesh Singhal, Cambridge University Press, 2008
2. Distributed Systems - Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, Prentice Hall of India, 2004

REFERENCES:

1. Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Hagit Attiya, Jennifer Welch, Wiley
2. Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen, Prentice Hall of India

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1IN401) COMMUNICATION TECHNOLOGIES FOR IOT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the five-layer internet protocol stack and MAC techniques
- To analyze physical layer operations, modulation, and duplexing methods
- To explore IEEE 802.11 WLAN standards, MAC, PHY, and security
- To study wireless PAN, MAN technologies, and security mechanisms
- To learn 6LoWPAN, bluetooth architectures, and emerging wireless technologies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Study different types of five-layer internet protocol stack and physical layer

CO-2: Understand the characteristics of WLAN, WPAN and their security issues

CO-3: Study various types of wireless MAN standards

CO-4: Comprehend the architecture and protocol stack of 6LoWPAN and Bluetooth Low Energy (BLE) technology

CO5: Demonstrate an understanding of emerging wireless technologies like 6LoWPAN, LoRa, and Bluetooth

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	-	2	3	-	-	1	2	-	2	2	1	2
CO-2	3	3	2	-	3	3	-	-	2	3	-	3	3	1	2
CO-3	3	3	3	2	-	3	-	-	2	3	-	2	3	2	3
CO-4	2	3	3	3	-	3	1	3	3	3	-	3	3	2	3
CO-5	3	3	3	3	3	3	-	3	3	3	-	3	3	3	3

UNIT-I:

Five Layer Internet Protocol Stack: Medium access control (MAC) layer-Aloha, CSMA/CA, TDMA, CDMA. Introduction to Cellular Technologies

UNIT-II:

Physical Layer (PHY) - Signal processing of bits and physical signals Modulation, Coding, Bit Interleaving, Synchronization, Carrier sensing and collision detection. Duplexing methods.

UNIT-III:

Wireless LAN: IEEE 802.11 WLAN standards, IEEE 802.11 MAC Layer, IEEE 802.11 PHYSICAL Layer, IEEE 802.11 Enhancement WLAN Security- Other WLAN Standards

UNIT-IV:

Wireless PAN and Wireless MAN: Introduction, IEEE 802.15.4 Wireless USB, ZigBee, IrDA, Wireless PAN Security.

IEEE 802.16 Wireless MAN Standard, Metropolitan Area, Mesh Network, Start-up phase and operating phase.

UNIT-V:

6LoWPAN and Bluetooth: Introduction, Architecture, Protocol stack of 6LoWPAN Architecture, Neighbor discovery, Challenges in 6LoWPAN. LoRa and LoRaWAN.

Introduction to Bluetooth, Network topology, BLE frequency channels, Message exchange, BLE protocol stack, Application.

TEXT BOOKS:

1. Wireless Networking Technology - From Principles to Successful Implementation, Steve Rackley, 1st Edition, Elsevier, 2007
2. Mobile Communication, Jochen Schiller, 2nd Edition, Pearson Education, 2010

REFERENCES:

1. The Internet of Things: Key Applications and Protocols, O. Hersent, 1st Edition, Wiley, 2013
2. Getting Started with Bluetooth Low Energy-Tools and Techniques of Low Power Networking, Akiba and Robert Davidson, 1st Edition, O'Reilly

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CS304) CYBER SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To summarize various types of cyber-attacks and cyber-crimes
- To understand cyber laws and the concepts of digital forensics
- To discuss safety measures for the protection of mobile and wireless devices
- To learn the organizational security implications and threats
- To study the impact of data privacy attacks on various domains

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the need of cyber security and various types of attacks

CO-2: Understand national and international regulations of cyber security and cyber forensics

CO-3: Interpret the security challenges related to mobile and wireless devices

CO-4: Analyze the security and privacy implications of an organization

CO-5: Examine the data privacy concepts and cybercrime in different domains

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	2	2	2	-	3	2	2	-	2	2	3	3
CO-2	1	1	3	3	1	1	-	-	3	3	-	2	1	3	3
CO-3	2	2	3	2	2	2	-	-	3	3	-	3	2	3	2
CO-4	2	2	3	2	3	2	-	3	3	3	2	3	2	2	3
CO-5	1	2	2	3	3	3	-	3	3	3	2	3	3	3	2

UNIT-I:

Introduction to Cyber Security: Basic Cyber Security Concepts, layers of security, Vulnerability, threat, Harmful acts, Internet Governance – Challenges and Constraints, Computer Criminals, CIA Triad, Assets and Threat, motive of attackers, active attacks, passive attacks, Software attacks, hardware attacks, Cyber Threats-Cyber Warfare, Cyber Crime, Cyber terrorism, Cyber Espionage, etc., Comprehensive Cyber Security Policy.

UNIT-II:

Cyberspace and the Law & Cyber Forensics: Introduction, Cyber Security Regulations, Roles of International Law. The INDIAN Cyberspace, National Cyber Security Policy, Historical background of Cyber forensics, Digital Forensics Science, The Need for Computer Forensics, Cyber Forensics and Digital evidence, Forensics Analysis of Email, Digital Forensics Lifecycle, Forensics Investigation, Challenges in Computer Forensics

UNIT-III:

Cybercrime: Mobile and Wireless Devices: Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, Organizational security Policies and Measures in Mobile Computing Era, Laptops.

UNIT-IV:

Cyber Security: Organizational Implications: Introduction, cost of cybercrimes and IPR issues, web threats for organizations, OWASP top 10 web application security risks, social media marketing: security risks and perils for organizations, social computing and the associated challenges for organizations

UNIT-V:

Privacy Issues: Basic Data Privacy Concepts: Fundamental Concepts, Data Privacy Attacks, Data linking and profiling, privacy policies and their specifications, privacy policy languages, privacy in different domains- medical, financial, etc Cybercrime: Examples and Mini-Cases Examples: Official Website of Maharashtra Government Hacked, Indian Banks Lose Millions of Rupees, Parliament Attack, Pune City Police Bust Nigerian Racket, e-mail spoofing instances. Mini Cases: The Indian Case of Online Gambling, An Indian Case of Intellectual Property Crime, Financial Frauds in Cyber Domain.

TEXT BOOKS:

1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole and Sunit Belpure, Wiley
2. Computer and Cyber Security: Principles, Algorithm, Applications, and Perspectives, B. B. Gupta, D. P. Agrawal, Haoxiang Wang, CRC Press

REFERENCES:

1. Mining of Massive Datasets, v2.1, Jure Leskovek, Anand Rajaraman, Jeffrey Ullman, Cambridge University Press, 2014
2. Data Science from Scratch: First Principles with Python, Joel Grus, 1st Edition, 2015
3. Doing Data Science, Straight Talk from the Frontline, Cathy O'Neil, Rachel Schutt, O' Reilly, 1st Edition, O'Reilly, 2013
4. Mining of Massive Datasets, Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman, 2nd Edition, Cambridge University Press, 2014

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1IN402) IOT FOR SMART CITIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand IoT-based smart city solutions, key technologies, and challenges
- To explore communication technologies and protocols for IoT through real-world use cases
- To analyze cybersecurity threats to medical IoT devices and security countermeasures
- To study intelligent transport systems and IoT-enabled electric vehicle management
- To examine IoT applications in water resource management and infrastructure monitoring

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concept of IoT in smart cities and explore IoT-based solutions for various smart city applications

CO-2: Learn communication technologies and protocols for IoT, with practical use-cases in intelligent transport systems and disaster management

CO-3: Analyze cyber security challenges and countermeasures for medical IoT devices in smart city healthcare services

CO-4: Explore intelligent transport systems, including smart vehicles, traffic management, and IoT-enabled electric vehicle charging solutions

CO-5: Study the management of water resources using IoT, focusing on smart water systems, irrigation, and leak detection technologies

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	-	3	3	2	2	2	3	-	3	3	2	2
CO-2	3	3	3	2	3	2	-	1	3	3	-	2	3	3	2
CO-3	3	3	2	3	3	3	1	2	2	3	-	3	2	3	2
CO-4	3	3	3	2	3	2	2	3	3	3	-	3	3	2	2
CO-5	3	3	3	2	3	3	2	1	2	3	-	3	3	1	3

UNIT-I:

IoT for Smart City - Overview and Key Challenges: Introduction to smart cities, characteristics of smart cities, IoT based solutions for smart cities-smart grid, smart home, smart health care, transport management, key enabling technologies for smart city, Challenges ahead.

UNIT-II:

Communication Technologies and Protocols for IoT: Introduction, communication technologies for IoT networks, recent protocols for IoT, study of communication technologies through use-cases: Intelligent transport system, disaster management.

UNIT-III:

Cyber Security Attacks on Medical IoT Devices For Smart City Health Care Services:

Smart city health care, IoT medical devices for smart city health care systems, cyber security attacks on medical IoT devices, security counter measurements.

UNIT-IV:

Intelligent Transport Systems: Smart vehicles and fuels, GIS, GPS, Navigation system, traffic safety management, mobility services, E-ticketing, smart parking system.

IoT-enabled Electric Vehicles in Smart Cities: introduction, EV charge scheduling and charging techniques, renewable energy for EV scheduling, smart distribution systems: smart EV scheduling case study.

UNIT-V:

Management of Water Resources and Related Infrastructure: Storage and conveyance system of water, sustainable water and sanitation, sewerage system, flood management, conservation systems. Smart water-challenges, IoT for smart billing for smart water, IoT in smart water irrigation, IoT in asset maintenance and leakage detection.

TEXT BOOKS:

1. The Age of Intelligent Cities: Smart Environments and Innovation-for-all Strategies (Regions and Cities), Nicos Komninos
2. Smart Cities: Big Data, Civic Hackers, and the Quest for a New Utopia, Anthony Townsend
3. Infrastructure engineering and management, Grig N. S., Wiley, 1988

REFERENCES:

1. Infrastructure Management, Hudson W. R., Haas R., Uddin W., McGraw-Hill, 1997
2. Mission Statement & Guidelines on Smart City Scheme, Government of India - Ministry of Urban Development
[http://smartcities.gov.in/upload/uploadfiles/files/Smart City Guidelines \(1\).pdf](http://smartcities.gov.in/upload/uploadfiles/files/Smart%20City%20Guidelines%20(1).pdf)
3. Smart City on Future Life - Scientific Planning and Construction, Xianyi Li
4. IoT for Smart Cities: Technologies, Big Data and Security-Waleem Ejad, Alagan Anapalagan
5. IoT Technologies In Smart Cities-From Sensors to Big Data, Security and Trust, Fadi Al-Turjaamn and Mohammed Imran

B.Tech. VII Semester

(22PC2IN401) EMBEDDED GPU LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: IoT Architecture, Python Programming

COURSE OBJECTIVES:

- To provide the fundamental concepts of database creation
- To implement the concepts of Data manipulation
- To develop procedures for querying Multiple tables
- To understand the concepts of PL / SQL
- To gain knowledge on CUDA programming constructs

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic programming and interfacing different I/Os

CO-2: Construct Embedded system application using Jetson Nano

CO-3: Apply CUDA for Kernel programming

CO-4: Design and implement algorithms using CUDA

CO-5: Develop and implement algorithms using OpenCL

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	3	2	2	1	2	-	1	-	-	1	2	2	3
CO-2	3	2	3	3	2	1	2	-	2	-	-	2	3	2	3
CO-3	3	2	3	3	2	2	2	-	2	-	-	2	3	2	3
CO-4	3	2	3	3	2	1	2	-	2	-	-	2	3	2	3
CO-5	3	2	3	3	2	2	2	-	3	3	2	2	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. GPIO programming (LED Blinking) on Jetson Nano.
2. GPIO programming for traffic light.
3. Interfacing sensors and actuators to Jetson Nano.
4. Interfacing camera and modules with Jetson Nano.
5. To perform data classification using Jetson Nano.
6. Write a CUDA program to demonstrate squaring an array using CUDA kernel.
7. Write a CUDA C program to add two large vectors.
8. Write a CUDA C program to calculate Matrix transpose and trace.
9. Design parallel algorithm for matrix multiplication using CUDA.
10. Write a CUDA programming for Application profiling using timers
11. Write a CUDA program to find out minimum among 100 values using a CUDA kernel.

12. Write an OpenCL program for matrix multiplication.
13. Write an OpenCL program for calculating matrix transpose and trace.
14. Write an OpenCL program for calculating value of π .

TEXT BOOKS:

1. Computer Architecture - A Quantitative Approach, John L. Hennessy and David A. Patterson, 5th Edition, Morgan Kaufmann.
2. Heterogeneous Computing with OpenCL, Benedict Gaster, Lee Howes, David R. Kaeli, Elsevier, 2013
3. OpenCL Programming Guide, Aaftab Munshi, Benedict Gaster, Timothy G. Mattson, James Fung & Dan Ginsburg, Addison-Wesley Professional, 2011
4. Jeston_TK1_User_Guide.pdf, NVIDIA

REFERENCES:

1. Programming Massively Parallel Processors-A hands-on-Approach, David Kirk, Wen-mei W. Hwu
2. Fundamentals of Parallel Multicore Architecture, Chapman and Hall/CRC Computational Science
3. Modern Processor Design: Fundamentals of Superscalar Processors, Shen John Paul, Lipasti, Mikko H.
4. General-Purpose Graphics Processor Architecture, Tor M. Aamodt, Wilson Wai Lun Fung, Timothy G. Rogers, Morgan and Claypool Publishers
5. <https://coreavi.com/wp-content/uploads/CoreAVI-White-Paper-Weighing-theFactors-of-Discrete-Versus-Embedded-GPUs.pdf>.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2AM402) DEVOPS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Proficiency in at least one programming language (e.g., Python, Java, or Shell scripting), Basic understanding of REST APIs and software development concepts

COURSE OBJECTIVES:

- To provide hands-on experience in developing backend applications, implementing CRUD operations, and exposing REST APIs
- To introduce version control systems using Git and GitHub, enabling effective source code management and collaboration
- To familiarize with continuous integration and deployment (CI/CD) pipelines, using tools like CircleCI and GitHub Actions
- To equip with containerization skills, enabling them to package and manage applications using Docker and Kubernetes
- To develop proficiency in API testing and automation, utilizing tools like Postman and writing test cases for backend services

COURSE OUTCOMES: After completion of the course, the student should be able to

CO1: Develop backend applications with CRUD operations and deploy them following DevOps best practices

CO2: Manage and collaborate on source code using Git and GitHub, demonstrating version control strategies

CO3: Set up and utilize CI/CD pipelines using CircleCI/GitHub Actions to automate builds and deployments

CO4: Containerize applications using Docker, automate deployments, and orchestrate them using Kubernetes

CO5: Perform API testing with Postman and write automated test cases to validate backend functionality

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	1	1	2	2	2	3	3	2	3
CO-2	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-3	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-4	3	3	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. Write code for a simple backend project. It should have basic CRUD operations and APIs (GET, POST, DELETE, PATCH/UPDATE).
2. Explore Git and GitHub commands.
3. Practice Source code management on GitHub. Experiment with the source code written in exercise 1.
4. Explore Docker commands for container management.
5. Develop a simple containerized application using Docker.
6. Integrate Kubernetes and Docker
7. Automate the process of running containerized application developed in exercise 7 using Kubernetes.
8. CircleCI/ Github Actions setup, explore the environment.
9. Demonstrate continuous integration and development using CircleCi/Github Actions
10. Install postman and explore manual API testing.
11. Develop test cases for simple APIs in the language of choice.
12. Integrate any of the suitable monitoring and logging dashboards with your application.
13. Hands-on parallel exploration of Terraform, and its configurations

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, 2018
2. DevOps Tools from Practitioner's Viewpoint, Deepak Gaikwad, Viral Thakkar, Wiley

REFERENCES:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley
2. Edureka DevOps Full Course - https://youtu.be/S_0q75eD8Yc

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PC1AM401) IMAGE PROCESSING AND COMPUTER VISION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basic knowledge of Mathematics, Including Linear Algebra, Calculus and Probability; Understanding of Signals, Fourier Transforms and Basic Algorithm

COURSE OBJECTIVES:

- To understand digital image fundamentals, including sampling, quantization, and color imaging
- To apply enhancement and morphological techniques in spatial and frequency domains
- To analyze image restoration, compression, and segmentation techniques
- To explore edge, blob, line, and corner detection algorithms for vision tasks
- To implement feature detection, matching, and applications like image stitching

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the general terminology of digital image processing

CO-2: Examine the image enhancement techniques in spatial & frequency domains

CO-3: Evaluate the methodologies for image segmentation, restoration

CO-4: Apply edge, line, and corner detection techniques to extract meaningful features from images

CO-5: Implement interest point detection, feature description, and matching methods for image analysis and stitching

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	1	1	1	1	1	1	1	1	2	3	2	1
CO-2	3	3	2	2	3	1	1	1	1	1	1	2	3	1	1
CO-3	3	3	2	3	1	1	1	1	1	1	1	2	3	2	3
CO-4	2	3	3	2	3	1	1	1	1	1	1	2	2	3	3
CO-5	2	2	2	2	3	1	1	1	1	1	1	3	3	3	3

UNIT-I:

Fundamentals of Image Processing: Digital Image Fundamentals, Basic steps of Image Processing System, Sampling and Quantization of an image, the relationship between pixels, Imaging Geometry. Color Imaging.

Image Transforms: DFT, DCT, Haar.

UNIT-II:

Image Enhancement: Spatial domain methods: Histogram processing, Fundamentals of Spatial filtering, smoothing spatial filters, Sharpening spatial filters.

Frequency Domain Methods: Basics of filtering in frequency domain, image smoothing, image sharpening, Selective filtering.

Morphological Image Processing: Dilation and Erosion, Opening and Closing, the hit or miss Transformation, and some basic morphological algorithms.

UNIT-III:

Image Restoration: Image Restoration Degradation model, Algebraic approach to restoration, Inverse Filtering, Least Mean square filters.

Image Compression: Image compression fundamentals - Coding Redundancy, Spatial and Temporal redundancy, Compression models: Lossy and Lossless, Huffman coding, Arithmetic coding, LZW coding, Run length coding, Bit plane coding, Transform coding, Predictive coding, JPEG Standards.

Image Segmentation: Segmentation concepts, Thresholding, Region Based segmentation.

UNIT-IV:

Introduction to Computer Vision and its Applications: A brief history, Edge Detection: Difference of Gaussian (DoG), Gradient magnitude, Canny Edge Detector; Blob Detector: Laplacian of Gaussian; Lines Detection: Hough transform algorithm; Corner Detection: Harris Corner Detector.

UNIT-V:

Interest Point Detection: Introduction

Feature Descriptors and matching: Histogram of Oriented Gradients (HoG), Local Binary Patterns (LBP), Scale Invariant Feature Transform (SIFT); Feature Matching: RANSAC algorithm, Application: Image Stitching; Visual Matching: Bag-of-words.

TEXT BOOKS:

1. Digital Image Processing, Rafael C. Gonzalez and Richard E. Woods, 3rd Edition, Pearson, 2008
2. Computer Vision: Algorithms and Applications, Richard Szeliski Springer, 2010 (<http://szeliski.org/Book/>)

REFERENCES:

1. Digital Image Processing, S. Sridhar, 2nd Edition, Oxford, 2016
2. Digital Image Processing, William K. Pratt, 3rd Edition, John Willey, 2004
3. Computer Vision: A Modern Approach, Forsyth and Ponce, Prentice Hall, 2002
4. Programming Computer Vision using Python (http://programmingcomputervision.com/downloads/ProgrammingComputerVision_CCdraft.pdf)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1IN403) WEARABLE COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of wearable sensors, their applications, challenges, and future trends
- To explore the role of IoT in wearable devices, smart connectivity, and real-world use cases
- To study energy harvesting techniques for powering wearable devices
- To analyze wireless health systems, Body Area Networks (BAN), and associated challenges
- To examine wearable system applications in healthcare, sports, and safety monitoring

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Comprehend the significance and role of for wearable system

CO-2: Illustrate appropriate sensors and signal processing techniques for wearable system

CO-3: Elucidate the level of energy involvement in wearable systems

CO-4: Outline the necessity of wireless health system

CO-5: Develop solutions for health monitoring and safety devices using wearable devices.

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	-	-	-	2	2	2	2	2	2	2
CO-2	3	2	2	2	3	-	-	-	2	2	2	2	2	2	2
CO-3	3	3	3	2	3	-	2	2	3	3	3	3	2	2	2
CO-4	3	3	3	3	3	-	2	2	3	3	3	3	2	2	2
CO-5	3	3	3	3	3	1	2	2	3	3	3	3	2	2	2

UNIT-I:

Basics of Sensors and Its Wearable Application: Role of Wearables, Attributes of Wearables, The Meta Wearables – Textiles and clothing, Social Aspects: Interpretation of Aesthetics, Adoption of Innovation, On- Body Interaction; Case Study: Google Glass, health monitoring, Wearables: Challenges and Opportunities, Future and Research Roadmap

UNIT-II:

Role of IOT in Wearable Devices: Smart connectivity and Big picture of IoT-smart devices, networks, Wireless technologies and need for data analysis. Evolution of

wearable technology, Wearable IoT use cases- Smart watches , Android wear, Smart glasses, fitness trackers, health care devices.

UNIT-III:

Energy Harvesting for Wearable Devices: Solar cell, Vibration based, Thermal based, Human body as a heat source for power generation, Hybrid thermoelectric photovoltaic energy harvests, Thermopiles.

UNIT-IV:

Wireless Health Systems: Need for wireless monitoring, Definition of Body area network, BAN and Healthcare, Technical Challenges- System security and reliability, BAN Architecture

UNIT-V:

Applications of Wearable Systems: Medical Diagnostics, Medical Monitoring-Patients with chronic disease, Hospital patients, Elderly patients, neural recording, Gait analysis, Sports Medicine, Safety devices.

TEXT BOOKS:

1. Wearable Sensors: Fundamentals, Implementation and Applications, Edward Sazonov, Michael R. Neuman, Elsevier, 2014
2. Wearable Electronics Sensors-For Safe and Healthy Living, Subhas C. Mukhopadhyay, Springer International Publishing, 2015
3. Body Area Networks Safety, Security, and Sustainability, Sandeep K. S. Gupta, Tridib Mukherjee, Krishna Kumar Venkata Subramanian, Cambridge University Press, 2013

REFERENCES:

1. Wearable Monitoring Systems, Annalisa Bonfiglio, Danilo De Rossi, Springer, 2011
2. Wearable eHealth systems for Personalized Health Management - State of the Art and Future Challenges, Andreas Lymberis, Danilo de Rossi, IOS Press, The Netherlands, 2004
3. Sensor Technology Hand Book, Jon. S. Wilson, Elsevier

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming and Data Structures

COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of augmented and virtual reality in interactive application development

CO-2: Apply Geometric Modeling Techniques

CO-3: Implement virtual environment with respect to user experience

CO-4: Design Virtual/Augmented Reality Applications

CO-5: Acquire fundamental and hands-on skills needed to develop AR/VR applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	1	2	2	1	-	2	2	3	3	3
CO-2	1	1	1	3	1	2	2	1	2	1	1	2	3	3	3
CO-3	1	1	2	-	3	1	1	1	1	2	1	2	3	3	3
CO-4	-	2	-	-	-	2	-	2	1	1	3	1	3	3	3
CO-5	-	-	2	-	3	-	-	-	-	-	-	-	3	3	3

UNIT-I:

Introduction to Augmented Reality (AR): Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

UNIT-II:

Introduction to Virtual Reality (VR): Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

UNIT-III:

Interaction Design for AR/VR Environments: Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

UNIT-IV:

Unity Overview: Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

UNIT-V:

Introduction to Vuforia and Physics in UNITY

Vuforia Overview: Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

Introduction to Scripting: Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

TEXT BOOKS:

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kaufman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

REFERENCES:

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

ONLINE RESOURCES:

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PC1CY401) BLOCKCHAIN TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

CO-2: Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

CO-3: Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

CO-4: Apply blockchain security principles and explore its applications, limitations, and future scope

CO-5: Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	3	1	1
CO-2	3	2	3	3	3	2	2	3	2	1	1	3	3	1	3
CO-3	2	2	2	1	1	1	1	2	1	3	1	2	1	3	1
CO-4	3	3	2	1	1	1	1	2	1	3	1	3	1	3	3
CO-5	3	3	3	1	1	1	1	1	1	3	1	2	1	3	1

UNIT-I:

Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

UNIT-II:

Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III:

Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

UNIT-IV:

Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

UNIT-V:

Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

TEXT BOOK:

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nifin Gaur, Pearson Addison Wesley

ONLINE RESOURCES:

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

B.Tech. VIII Semester

(22PE1IN404) IOT SYSTEM DESIGN

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To familiarize with various models and design aspects of IoT system
- To demonstrate modelling of IoT systems
- To design and build IoT Data system
- To understand modelling of IoT systems using UML diagrams
- To analyze the usage of DevOps in IoT systems design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the components associated complexity of IoT system design

CO-2: Model real-time and IoT systems from specifications

CO-3: Design and implement hardware and software for a IoT application

CO-4: Apply UML diagrams to model IoT systems in real time

CO-5: Use DevOps technologies for IoT System development, testing and operation

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	-	1	-	-	1	-	-	1	1	-	1	1	2	1
CO-2	1	-	-	-	2	-	-	-	-	1	-	-	1	1	2
CO-3	1	-	2	-	2	1	-	-	2	-	-	1	1	2	2
CO-4	1	-	2	2	1	-	-	-	-	2	-	-	1	2	2
CO-5	1	-	2	-	-	-	-	-	-	1	-	1	1	1	2

UNIT-I:

IoT Systems: A Reconnaissance flyover, IoT Systems functional view, infrastructure view, IoT project design and implementation, Types of IoT Projects, IoT system design, System implementation, system integration, Deployment, System case study (POEM) Personal office energy monitor. Introduction to IoT Architecture — Fog, Edge and Cloud in IoT

UNIT-II:

Hardware Design Fundamentals: Introduction, Hardware System Requirements, Hardware Functional Specifications, Software Functional Specifications, Hardware Component Selection IoT, Design for Manufacturability, Design for Testability, Schematics, Layouts and Gerbers, PCB Fabrication and Assembly, Summarising hardware design flow

Use Case: Applying learning to Connected vehicles, Examples of Commercially available IoT Devices, Standards used in IoT component Configurations, Selection of

Network Hardware for IoT Applications, IoT Hardware Security, Energy Management in IoT, Energy harvesting, Energy harvesting schemes.

UNIT-III:

IoT Data Systems Design: Introduction, IoT Data System, Value Chain Activity Generates Data, Operational Data: Sensors and Devices, Business and Customer/User Data, Intersection of Operational Data and Business Data, Structured and Unstructured Data, Data Systems Design, Data Transport, Data Collection, Data Storage, Data Preparation, Organizing Data for Analytics, Data Lake, Data Warehouses, and Efficient Access to Data, Analytics and Business Intelligence, Data Science and IoT, Machine Learning, Putting It Together to Build an IoT Data System, Applying Learnings to Connected Vehicle Use Case.

UNIT-IV:

IoT system Development Methods: Introduction, System Development Methods, IoT SDMs in the Literature, Evaluation of IoT SDMs.

Modelling and Designing of IoT systems using UML Diagrams: An Introduction, the need, modelling of IoT systems using UML diagrams, architectural design of any IoT system to behavioral aspects.

UNIT-V:

New Process Model in IoT based Software Engineering: IoT based Software Development Life Cycle (SDLC), identifying business needs through IoT, Integrating Software Engineering practices in to IoT System Implementation: Devops, Devops Tool chain, DevSecops, Building DevOps culture.

TEXT BOOKS:

1. Design of Secure IoT Systems: A Practical Approach Across Industries, Sumeet Arora, Ramachandra Gambheer, Meenakshi Vohra, McGraw-Hill Education, 2021
2. Integrating the Internet of Things into Software Engineering Practices, D. Jeya Mala, 2019

REFERENCES:

1. Internet of Things: Concepts and System Design, Milan Milenkovic, Springer, 2020
2. Internet of Things: Challenges, Advances, and Applications, Edited by Qusay F. Hassan, Atta ur Rehman Khan, Sajjad A. Madani, Chapman and Hall/CRC Press, 2020
3. Energy Conservation for IoT Devices Concepts, Paradigms and Solutions, Mittal M., Tanwar S., Agarwal B., Goyal L. M. (Eds.), Springer, 2019
4. Fundamentals of IoT and Wearable Technology Design, Haider Raad, Wiley-IEEE Press, 2020
5. Internet of Things: Concepts and System Design, Milan Milenkovic, Springer, 2020

B.Tech. VIII Semester

(22PE1IN405) 5G AND IOT TECHNOLOGIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To familiarize with various wireless communications
- To understand channel modeling requirements
- To infer 5G for the device to device application
- To analyze IoT security using 5G technology
- To apply skills required to integrate IoT and 5G for smart applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the overview of 5G requirements

CO-2: Realize the revolution of Internet in mobile devices, cloud & sensor networks

CO-3: Interpret and illustrate device to device application using 5G

CO-4: Demonstration of security of IoT using 5G

CO-5: Apply skills required to integrate IoT and 5G for smart applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	1	1	1	-	-	1	2	-	1	2	1	1
CO-2	2	-	1	2	1	2	-	-	1	1	-	2	2	1	1
CO-3	1	2	-	-	-	1	-	-	2	1	-	1	1	1	2
CO-4	1	2	1	2	2	1	-	-	2	1	-	-	2	1	1
CO-5	2	2	1	2	1	1	-	-	1	2	-	1	1	1	1

UNIT-I:

Overview of 5G Broadband Wireless Communications: Evolution of mobile technologies 1G to 4G (LTE, LTEA, LTEA Pro), An Overview of 5G requirements, Regulations for 5G, Spectrum Analysis and Sharing for 5G.

UNIT-II:

The 5G Wireless Propagation Channels: Channel modeling requirements, propagation scenarios and challenges in the 5G modeling, Channel Models for Wave, MIMO Systems, 3G PP standards for 5G, IEEE 802.15.4

UNIT-III:

Device to Device Application: D2D from 4G to 5G, D2D standardization: 4G LTE D2D, D2D in 5G: research challenges, Radio resource management for mobile broadband D2D, RRM techniques for mobile broadband D2D, RRM and system design for D2D, 5G D2D RRM concept: an example, Multi-hop D2D communications for proximity and

emergency, services, National security and public safety requirements in 3GPP and METIS, Device discovery without and with network assistance

UNIT-IV:

IoT & 5G Security: IoT Advance Wireless Interfaces, Security challenges in IoT over 5G networks, Role of 5G in improving IoT security, Zero Trust Architecture in IoT

UNIT-V:

IoT Physical Devices and Endpoints - IoT and 5G Integration -How 5G Enhances IoT Applications - Low Latency and High Throughput for IoT-5G as the Backbone for Smart Cities, Autonomous Vehicles, and Healthcare, Overview of 6G

TEXT BOOKS:

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015
2. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014

REFERENCES:

1. Fundamentals of 5G Mobile Networks, Jonathan Rodriguez, John Wiley & Sons
2. Essentials of LTE and LTE-A, Amitabha Ghosh and Rapeepat Ratasuk, Cambridge University Press
3. New Directions in Wireless Communication Systems from Mobile to 5G, Athanasios G. Kanatos, Konstantina S. Nikita, Panagiotis Mathiopoulos, CRC Press
4. Millimeter Wave Wireless Communications, Theodore S. Rappaport, Robert W. Heath, Robert C. Danials, James N. Murdock, Prentice Hall Communications

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B.Tech. VIII Semester

(22PE1IN406) INDUSTRIAL IOT 4.0

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To develop Knowledge of basics, systems and enablers of Industry 4.0
- To understand the technical complexity of IIoT and Industry 4.0
- To impart the knowledge on framework for deploying an IIoT systems
- To explore protocols related to IIoT that enable to meet Industry 4.0 standards
- To explore the concepts and applications of Industry 4.0 and IoT technologies in smart manufacturing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Establish knowledge in a concise manner about IIoT ecosystem

CO-2: Identify the architectural models of various IIoT systems and the role of internet systems

CO-3: Understand and use various protocols of IIoT systems along with security aspects

CO-4: Introduce the Industry 4.0 along with exploration of some case studies.

CO-5: Evaluate the security considerations and challenges associated with IoT-based systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	3	1	1	-	-	-	-	-	1	2	2	2
CO-2	3	3	3	3	1	1	-	-	-	-	-	1	2	2	2
CO-3	3	2	3	3	1	1	-	-	-	-	-	1	2	2	2
CO-4	3	2	2	2	1	1	-	-	-	-	-	1	2	2	2
CO-5	3	3	3	3	1	2	-	-	-	-	2	1	2	2	2

UNIT-I:

Introduction: Industrial Internet, Key IIoT Technologies, Intelligent Devices, Opportunities and Benefits, The digital and Human workforce.

Industrial Internet Use-Cases: Healthcare, Oil & Gas Industry, Smart Office, Logistics and the Industrial Internet, Retail, IoT Innovations in Retail.

UNIT-II:

The Technical and Business Innovators of the Industrial Internet: Miniaturization, Cyber Physical Systems, Wireless Technology, IP Mobility, Network Functionality Virtualization, Software Defined Networks, Smart phones, The cloud and Fog, Big Data and Analytics, M2M Learning and AI, Augmented Reality, 3D Printing, People versus Automation.

UNIT-III:

IIoT Reference Architecture: The IIC Industrial Internet Reference Architecture, Industrial Internet Architecture Framework, Architectural Topology, Three-Tier Topology, Connectivity, Key System Characteristics, Data management. Designing Industrial Internet Systems: The concept of the IIoT. Middleware Industrial Internet of Things Platforms: Introduction, Middleware Architecture.

UNIT-IV:

IIoT WAN Technologies and Protocols: Introduction, IIoT Device Low-Power WAN Optimized Technologies for M2M(SigFox, LoRaWAN, nWave, RPMA, Low Power Wi-Fi, LTE Category-M, Millimeter Radio. Securing the Industrial Internet: Introduction, Security in Manufacturing, PLCs and DCS, Securing the OT, Network level and System Level, Identity Access Management

UNIT-V:

Introducing Industry 4.0: Introduction, Defining Industry 4.0, Characteristics of Industry 4.0, The Value Chain, Differential Perspective, Benefits to Business, Industry 4.0 Design Principles, Building Blocks of Industry 4.0....

Smart Factories: Introduction, Smart Factories in Action, Real-World Smart Factories(GE's Brilliant Factory, Airbus, Siemens' Amberg Electronics Plant), Industry 4.0: The way forward

TEXT BOOK:

1. Industry 4.0: The Industrial Internet of Things, Alasdair Gilchrist, Apress, 2017

REFERENCES:

1. Industrial Internet of Things: Cyber manufacturing Systems, Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat, Springer, 2017
2. Hands-On Industrial Internet of Things: Create a Powerful Industrial IoT, Giacomo Veneri, Antonio Capasso, Packt, 2018

B.Tech. VIII Semester

(22PE1DS406) QUANTUM COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore the quantum computation environment

CO-2: Identify the implications of quantum computation in the field of mathematics and associated fields

CO-3: Demonstrate building blocks and programming methodologies for quantum computing

CO-4: Apply quantum computation algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	1	1	1	1	1	-	-	-	1	1	-	1
CO-2	2	3	3	2	1	1	1	1	1	-	1	1	2	-	1
CO-3	2	2	2	1	1	1	1	1	-	-	1	1	1	1	1
CO-4	2	2	1	1	1	1	1	1	-	-	-	1	1	-	1
CO-5	1	3	3	2	2	1	1	1	1	-	1	1	2	1	1

UNIT-I:

Introduction and Background: Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

Linear Algebra and The Dirac Notation: The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

UNIT-II:

Qubits and The Framework of Quantum Mechanics: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

UNIT-III:

Superdense Coding and Quantum Teleportation: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.

Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

UNIT-IV:

Programming Languages: Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

Theoretical Computer Science: Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

UNIT-V:

Quantum Algorithms: Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

TEXTBOOKS:

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

REFERENCES:

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom

B.Tech. VIII Semester

(22PE1IN407) FOG AND EDGE COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Machine Learning, IoT Architecture and Protocols

COURSE OBJECTIVES:

- To explore the basics of new computing paradigms
- To familiarize with the concepts of IoT, Fog and Cloud Computing
- To expose the middleware for edge cloud architectures
- To exploiting Fog Computing case studies and testing perspectives
- To realizing the data analytics concepts

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze the basics of new computing paradigms

CO-2: Understand the concepts of IoT, Fog and Cloud Computing

CO-3: Examine the middleware for Edge Cloud Architectures

CO-4: Develop Fog Computing case studies with testing perspectives

CO-5: Realizing the data analytics concepts

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	3		-	-	-	-	2	2	2	2
CO-2	3	2	1	2	1	2	-	-	-		-	3	2	2	2
CO-3	2	2	2	2	2	2	-	-	-	-	-	2	2	2	2
CO-4	3	2	2	2	2	1	-	-	-	-	-	2	2	2	2
CO-5	1	2	1	2	1	1	-	-	-	-	-	3	2	2	3

UNIT-I:

Internet of Things (IoT) and New Computing Paradigms: Introduction, Relevant, Technologies, Fog and Edge Computing Completing the Cloud, Hierarchy of Fog and Edge Computing, Business Models, Opportunities and Challenges.

Addressing the Challenges in Federating Edge Resources: Introduction, The Networking Challenge, The Management Challenge, Miscellaneous Challenges

UNIT-II:

Integrating IoT + Fog + Cloud Infrastructures: System Modeling and Research Challenges, Introduction, Methodology, Integrated C2F2T Literature by Modeling Technique, Integrated C2F2T Literature by Use-Case Scenarios, Integrated C2F2T Literature by Metrics, Future Research Directions Optimization Problems in Fog and Edge Computing: Introduction, Background / Related Work, Preliminaries, The Case for Optimization in Fog Computing, Formal Modeling Framework for Fog Computing,

Metrics, Optimization Opportunities along the Fog Architecture, Optimization Opportunities along the Service Life Cycle.

UNIT-III:

Middleware for Fog and Edge Computing: Design Issues, Introduction, Need for Fog and Edge Computing Middleware, Design Goals, State-of-the-Art Middleware Infrastructures, System Model, Proposed Architecture.

A Lightweight Container Middleware for Edge Cloud Architectures: Introduction, Background/Related Work, Clusters for Lightweight Edge Clouds, Architecture Management – Storage and Orchestration, IoT Integration.

UNIT-IV:

Using Machine Learning for Protecting the Security and Privacy of Internet of Things (IoT) Systems: Introduction, Background, Survey of ML Techniques for Defending IoT Devices, Machine Learning in Fog Computing.

Fog Computing Realization for Big Data Analytics: Introduction, Big Data Analytics, Data Analytics in the Fog, Prototypes and Evaluation.

UNIT-V:

Exploiting Fog Computing in Health Monitoring: Introduction, An Architecture of a Health Monitoring IoT-Based System with Fog Computing, Fog Computing Services in Smart E-Health Gateways, System Implementation, Case Studies, Experimental Results, and Evaluation.

Testing Perspectives of Fog-Based IoT Applications: Introduction, Background, Testing Perspectives.

Legal Aspects of Operating IoT Applications in the Fog: Introduction, Related Work Classification of Fog/Edge/IoT Applications,

TEXT BOOKS:

1. Fog and Edge Computing: Principles and Paradigms, Rajkumar Buyya and Satish Narayana Srirama, Wiley Series on Parallel and Distributed Computing
2. IoT and Edge Computing for Architects Implementing Edge and IoT Systems From Sensors To Clouds With Communication Systems, Analytics and Security, Perry Lea, 2nd Edition
3. Edge Computing: A Primer, Jie Cao, Quan Zhang, Weisong Shi
4. Fog Computing: Theory and Practice, Assad Abbas, Samee U. Khan, Albert Y. Zomaya

REFERENCES:

1. Understanding the Infrastructure Edge Computing, Alex Marcham
2. Edge Computing and IoT: Systems, Management and Security, First International Conference, ICECI 2020
3. Sensors, Cloud, and Fog: The Enabling Technologies for the Internet of Things, Sudip Misra, Subhadeep Sarkar, Subarna Chatterjee
4. Fog Computing and Its Role in the Internet of Things, Flavio Bonomi, Rodolfo Milito, Jiang Zhu, Sateesh Addepalli, MCC'12, 2012, Helsinki, Finland

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1IN408) PRIVACY AND SECURITY IN IOT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the fundamentals, various attacks and importance of Security aspects in IoT
- To understand the techniques, protocols and some idea on security towards gaming models
- To understand the operations of Bitcoin blockchain, crypto-currency as application of blockchain technology
- To understand the essential components of IoT
- To understand security and privacy challenges of IoT

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Incorporate the best practices learnt to identify the attacks and mitigate the same

CO-2: Adopt the right security techniques and protocols during the design of IoT products

CO-3: Assimilate and apply the skills learnt on ciphers and block chains when appropriate

CO-4: Describe the essential components of IoT

CO-5: Find appropriate security/privacy solutions for IoT

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	1	-	-	-	-	-	2	2	2	2
CO-2	3	2	1	2	1	2	-	-	-	-	-	3	2	2	2
CO-3	3	2	2	1	2	2	-	-	-	-	-	2	2	2	2
CO-4	3	2	2	1	2	1	-	-	-	-	-	2	2	2	2
CO-5	1	2	1	2	1	1	-	-	-	-	2	3	2	2	2

UNIT-I:

Fundamentals of IoT and Security and its need, Prevent Unauthorized Access to Sensor Data, Block ciphers, Introduction to Blockchain, Introduction of IoT devices, IoT Security Requirements, M2M Security, Message integrity, Modeling faults and adversaries, Difference among IoT devices, computers, and embedded devices.

UNIT-II:

IoT and cyber-physical systems RFID Security, Authenticated encryption Byzantine Generals problem sensors and actuators in IoT. IoT security (vulnerabilities, attacks,

and countermeasures), Cyber Physical Object Security, Hash functions, Consensus algorithms and their scalability problems, Accelerometer, photoresistor, buttons.

UNIT-III:

Security engineering for IoT development Hardware Security, Merkle trees and Elliptic curves digital signatures, verifiable random functions, Zero-knowledge systems motor, LED, vibrator. IoT security lifecycle, Front-end System Privacy Protection, Management, Secure IoT Databases, Public-key crypto (PKI), blockchain, the challenges, and solutions, analog signal vs. digital signal.

UNIT-IV:

Data Privacy Networking Function Security Trees signature algorithms proof of work, Proof of stake, Networking in IoT, Device/User Authentication in IoT IoT Networking Protocols, Crypto-currencies, alternatives to Bitcoin consensus, Bitcoin scripting language and their use Real-time communication.

UNIT-V:

Introduction to Authentication Techniques Secure IoT Lower Layers, Bitcoin P2P network, Ethereum and Smart Contracts, Bandwidth efficiency, Data Trustworthiness in IoT Secure IoT Higher Layers, Distributed consensus, Smart Contract Languages and verification challenges data analytics in IoT - simple data analyzing methods.

TEXT BOOKS:

1. Practical Internet of Things Security, B. Russell and D. Van Duren, Packt Publishing, 2016
2. Security and Privacy in Internet of Things (IoT): Models, Algorithms, and Implementations, FeiHU, CRC Press, 2016
3. Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Narayanan et al., Princeton University Press, 2016

REFERENCES:

1. Mastering Bitcoin: Unlocking Digital Cryptocurrencies, A. Antonopoulos, O'Reilly, 2014
2. Network Security: A Decision and Game-theoretic Approach, T. Alpcan and T. Basar, Cambridge University Press, 2011
3. Security and the IoT ecosystem, KPMG International, 2015
4. Internet of Things: IoT Governance, Privacy and Security Issues, European Research Cluster
5. Security of Things: An Implementers' Guide to Cyber-Security for Internet of Things Devices and Beyond, Ollie Whitehouse, NCC Group, 2014